

Generalized Linear Mixed Models

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What you should be able to do after this class

1. Recognise when observations are clustered.
2. Specify random-intercept and random-slope structures.
3. Distinguish fixed coefficients from cluster-specific random deviations.
4. Fit and diagnose a GLMM in R.
5. Distinguish cluster-specific from population-averaged quantities.
6. Report estimates without overstating what a cross-sectional analysis can show.

Take-home: The goal is not only to fit a model, but to align the model, estimand and interpretation.

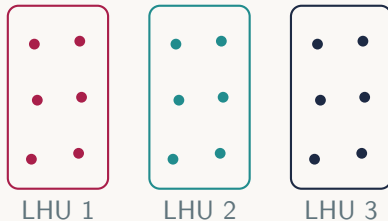
1. Why multilevel?

Dependence, pooling and what a single-level model misses

The unit of analysis is not always the unit of independence

In PASSI, people are sampled within local health units (LHUs). People from the same LHU may share:

- service organisation and prevention policies;
- local socioeconomic and cultural context;
- interviewer or fieldwork processes;
- unmeasured contextual determinants.



Take-home: Ignoring within-LHU dependence can make standard errors too small and hides between-LHU heterogeneity.

Three strategies: complete, no and partial pooling

Strategy	Assumption	Main consequence
Complete pooling	All LHUs share one intercept	Stable but ignores contextual heterogeneity
No pooling	One unrelated intercept per LHU	Flexible but noisy for small clusters
Partial pooling	LHU intercepts arise from a common distribution	Data-rich LHUs speak more; sparse LHUs shrink toward the mean

no pooling partial pooling complete pooling

→ less ← pooling → more

From a GLM to a GLMM

Let Y_{ij} denote observation i in cluster j , with conditional mean $\mu_{ij} = E(Y_{ij} \mid \mathbf{b}_j)$.

Generalized linear model

$$g(\mu_{ij}) = \alpha + \mathbf{x}_{ij}^\top \boldsymbol{\beta}$$

The same regression surface applies to every observation.

Generalized linear mixed model

$$g(\mu_{ij}) = \alpha + \mathbf{x}_{ij}^\top \boldsymbol{\beta} + \mathbf{z}_{ij}^\top \mathbf{b}_j$$

The random term allows the regression surface to vary across clusters.

2. GLMMs

Outcome distribution, link function, fixed effects and random effects

The general GLMM (Generalized linear mixed models)

Conditionally on the random effects,

$$Y_{ij} \mid \mathbf{b}_j \sim \mathcal{F}(\mu_{ij}, \phi), \quad g(\mu_{ij}) = \eta_{ij} = \alpha + \mathbf{x}_{ij}^\top \boldsymbol{\beta} + \mathbf{z}_{ij}^\top \mathbf{b}_j.$$

Distribution

$\mathcal{F}$ describes the conditional outcome distribution.

Link function

$g(\cdot)$ connects the mean to an unrestricted scale.

Fixed part

α and $\boldsymbol{\beta}$ describe systematic associations.

Random part

$\mathbf{b}_j$ represents cluster-specific deviations.

Take-home: A GLMM is a GLM whose linear predictor also contains random effects.

Two equivalent random-intercept parameterizations

For one predictor x_{ij} , the model may be written as

$$g(\mu_{ij}) = u_j + \beta x_{ij}, \quad u_j \sim N(\alpha, \tau^2),$$

or equivalently as

$$g(\mu_{ij}) = \alpha + \beta x_{ij} + b_j, \quad b_j \sim N(0, \tau^2).$$

$$u_j = \alpha + b_j$$

- u_j is the complete intercept for cluster j .
- α is the mean intercept across the cluster distribution.
- b_j is the deviation of cluster j from that mean.

Both formulations imply the same fitted model. The second makes the fixed and random components explicit.

What do α , β , $\mathbf{b}_j$ and τ^2 mean?

Term	Statistical role	Interpretation
α	fixed intercept	expected value of the linear predictor when covariates equal zero or their reference values and $\mathbf{b}_j = \mathbf{0}$
β_k	fixed slope	conditional change in the linear predictor for a one-unit increase in x_k , holding the other predictors and random effects fixed
b_j	random intercept	deviation of cluster j from the mean intercept α
τ^2	variance component	residual between-cluster heterogeneity on the link scale
ϕ	dispersion parameter	additional conditional variability when the selected family estimates one

A meaningful interpretation of α requires meaningful zero values and reference categories.

Fixed effects and random effects

For one continuous predictor x_{ij} , the most general model considered here is

$$\eta_{ij} = \text{logit}(p_{ij}) = \underbrace{\alpha + \beta x_{ij}}_{\text{fixed part}} + \underbrace{b_{0j} + b_{1j} x_{ij}}_{\text{random part}}.$$

Quantity	Role	Interpretation
α	fixed intercept	mean intercept across the distribution of LHUs
β	fixed slope	mean within-LHU change in log-odds for one unit of x
b_{0j}	random intercept deviation	difference between the intercept of LHU j and α
b_{1j}	random slope deviation	difference between the slope of LHU j and β

Take-home: For LHU j , the intercept is $\alpha + b_{0j}$ and the slope is $\beta + b_{1j}$.

Parameters and cluster-specific realizations

Fixed effects

α and β are unknown constants estimated once for the population of LHUs.

They describe the centre of the random-effects distribution.

Random effects

b_{0j} and b_{1j} are unobserved LHU-specific realizations from a probability distribution.

The model predicts them with shrinkage after estimating their variance.

$$E(b_{0j}) = E(b_{1j}) = 0.$$

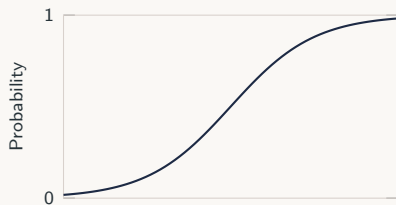
The word “random” refers to the model for differences among LHUs. It does not mean that the estimated effect changes at random each time the analysis runs.

Four random-effects structures

Model	Linear predictor	Intercept in LHU j	Slope in LHU j
Fixed intercept and slope	$\alpha + \beta x_{ij}$	α	β
Random intercept	$\alpha + \beta x_{ij} + b_{0j}$	$\alpha + b_{0j}$	β
Random slope only	$\alpha + \beta x_{ij} + b_{1j} x_{ij}$	α	$\beta + b_{1j}$
Random intercept and slope	$\alpha + \beta x_{ij} + b_{0j} + b_{1j} x_{ij}$	$\alpha + b_{0j}$	$\beta + b_{1j}$

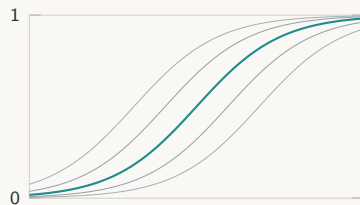
- A random intercept allows different baselines while retaining one common association with x .
- A random slope allows the association with x to vary across LHUs.
- The combined model allows both quantities to vary and estimates their covariance.

Even more complex



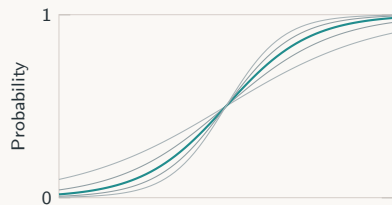
Fixed intercept + fixed slope

$$\eta_{ij} = \alpha + \beta x_{ij}$$



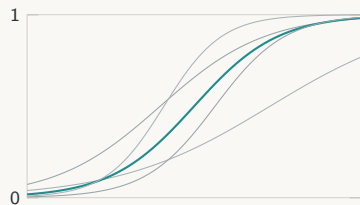
Random intercept

$$\eta_{ij} = \alpha + \beta x_{ij} + b_{0j}$$



Random slope only

$$\eta_{ij} = \alpha + (\beta + b_{1j})x_{ij}$$



Random intercept + random slope

$$\eta_{ij} = (\alpha + b_{0j}) + (\beta + b_{1j})x_{ij}$$

Random intercept: different baselines, common slope

$$\text{logit}(p_{ij}) = \alpha + \beta x_{ij} + b_{0j}, \quad b_{0j} \sim N(0, \tau_0^2).$$

Across LHUs

The baseline log-odds change by b_{0j} .
At $x = 0$, LHU j has odds

$$\exp(\alpha + b_{0j}).$$

Thus $\exp(b_{0j})$ is its baseline odds multiplier.

Within an LHU

For a one-unit change in x ,

$$\log(\text{OR}_j) = \beta.$$

The term b_{0j} cancels, so every LHU has the same conditional odds ratio $\exp(\beta)$.

The curves are parallel on the logit scale. Their separation on the probability scale varies with x because the inverse logit is nonlinear.

Random slope: common baseline, different associations

$$\text{logit}(p_{ij}) = \alpha + \beta x_{ij} + b_{1j}x_{ij}, \quad b_{1j} \sim N(0, \tau_1^2).$$

At $x = 0$, every LHU has the same intercept α because the random term $b_{1j}x$ equals zero. The curves therefore meet at the centred value $x = 0$.

Within LHU j ,

$$\log(\text{OR}_j) = \beta + b_{1j}, \quad \text{OR}_j = \exp(\beta + b_{1j}).$$

Different LHUs can have weaker, stronger or oppositely directed associations.

Caution: A slope-only random structure imposes zero residual baseline heterogeneity at $x = 0$. This restriction often lacks substantive support.

Random intercept and random slope

If the association with age varies across LHUs, write

$$\text{logit}(p_{ij}) = \alpha + \beta^\top \mathbf{x}_{ij} + b_{0j} + b_{1j}\text{Age10}_{ij},$$

with

$$\begin{pmatrix} b_{0j} \\ b_{1j} \end{pmatrix} \sim N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{01} & \tau_1^2 \end{pmatrix} \right].$$

- τ_0^2 measures residual variation in LHU intercepts.
- τ_1^2 measures variation in age slopes across LHUs.
- τ_{01} describes whether higher-baseline LHUs tend to have steeper or flatter age gradients.

For LHU j , the baseline log-odds are $\alpha + b_{0j}$ and the age log-odds ratio is $\beta_{\text{age}} + b_{1j}$.

The family and link define the scale of interpretation

Outcome	Family	Common link	Interpretation of η_{ij}
Continuous	Gaussian	identity	conditional mean
Binary	Binomial	logit	conditional log-odds
Count	Poisson	log	log conditional rate or mean count
Overdispersed count	Negative binomial	log	log conditional mean count
Proportion with denominator	Binomial	logit	conditional log-odds of success

$$\mu_{ij} = g^{-1}(\eta_{ij}).$$

Take-home: The same linear predictor has a different substantive meaning under a different link.

The fixed effects live on the link scale

For two observations in the same cluster that differ by one unit in x_k ,

$$\eta(\mathbf{x} + \mathbf{e}_k, \mathbf{b}_j) - \eta(\mathbf{x}, \mathbf{b}_j) = \beta_k.$$

- With the identity link, β_k is a difference in conditional means.
- With the log link, $\exp(\beta_k)$ is a conditional rate or mean ratio.
- With the logit link, $\exp(\beta_k)$ is a conditional odds ratio.

Caution: A coefficient must be interpreted on the scale implied by the link function.

Probability, odds and log-odds

For a binary outcome with probability $p = P(Y = 1)$,

$$\text{odds} = \frac{p}{1-p}, \quad \text{logit}(p) = \log\left(\frac{p}{1-p}\right).$$

p	$1-p$	odds	log-odds
0.20	0.80	0.25	-1.386
0.50	0.50	1.00	0
0.80	0.20	4.00	1.386

Odds compare the probability of the event with the probability of no event.

The logit maps probabilities in $(0, 1)$ to the whole real line.

$$p = \text{logit}^{-1}(\eta) = \frac{\exp(\eta)}{1 + \exp(\eta)}.$$

The intercept α in a logistic GLMM

Consider

$$\text{logit}(p_{ij}) = \alpha + \beta x_{ij} + b_j.$$

When $x_{ij} = 0$ and $b_j = 0$,

$$\alpha = \log\left(\frac{p_0}{1 - p_0}\right), \quad \exp(\alpha) = \frac{p_0}{1 - p_0}, \quad p_0 = \frac{\exp(\alpha)}{1 + \exp(\alpha)}.$$

- α is the baseline log-odds for a cluster at the centre of the random-effect distribution (i.e. random effect equal to zero).
- $\alpha + b_j$ is the baseline log-odds for cluster j .
- $\exp(b_j)$ multiplies the baseline odds for cluster j .

From β to an odds ratio

For a binary predictor $x_{ij} \in \{0, 1\}$ in the same cluster,

$$\text{logit}\{p_j(1)\} = \alpha + \beta + b_j,$$

$$\text{logit}\{p_j(0)\} = \alpha + b_j.$$

Therefore,

$$\log(\text{OR}) = \beta, \quad \text{OR} = \exp(\beta).$$

β	$\exp(\beta)$	Interpretation
0	1.00	equal conditional odds
0.405	1.50	50% higher conditional odds
-0.693	0.50	50% lower conditional odds

The comparison must hold the random effect fixed

Within the same cluster j ,

$$\text{logit}\{p_j(1)\} - \text{logit}\{p_j(0)\} = \beta.$$

Across two clusters $j \neq j'$,

$$\text{logit}\{p_j(1)\} - \text{logit}\{p_{j'}(0)\} = \beta + (b_j - b_{j'}).$$

- $\exp(\beta)$ compares observations with the same random-effect value.
- The GLMM coefficient is therefore called subject-specific or cluster-specific.
- A population-averaged comparison requires integration over the random effects.

Take-home: Holding the cluster effect fixed is part of the estimand.

3. The logistic GLMM

Conditional associations, heterogeneity and prediction

One equation, three ingredients (random intercept)

$$\underbrace{\text{logit}\{P(Y_{ij} = 1 \mid b_j)\}}_{\text{link + conditional mean}} = \underbrace{\alpha + \mathbf{x}_{ij}^\top \boldsymbol{\beta}}_{\text{fixed part}} + \underbrace{b_j}_{\text{random part}}$$

Outcome model

Bernoulli response:
 $Y_{ij} \mid b_j \sim \text{Bernoulli}(p_{ij})$.

Linear predictor

Covariates explain systematic individual differences.

Random effect

$b_j \sim N(0, \tau^2)$ allows baseline risk to vary across LHUs.

Conditional independence is assumed once covariates and b_j are known.

Fixed effects are cluster-specific in a logistic GLMM

For a one-unit increase in x_k , holding other covariates and the same LHU effect fixed:

$$\exp(\beta_k) = \frac{\text{odds at } x_k + 1 \text{ in LHU } j}{\text{odds at } x_k \text{ in the same LHU } j}.$$

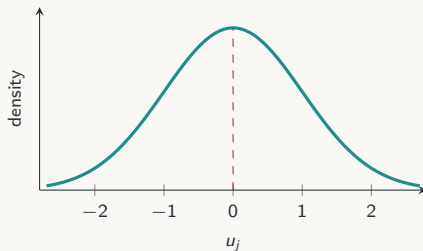
- This is a **subject-/cluster-specific odds ratio**.
- It compares otherwise similar people within the same latent context.
- It is generally not equal to a population-averaged odds ratio because the logit link is nonlinear.

Caution: Do not describe $\exp(\beta_k)$ as a risk ratio, and do not silently switch from conditional to marginal language.

The random intercept changes baseline odds

At fixed covariates, LHU j has odds multiplier $\exp(b_j)$ relative to a cluster at the centre of the random-effect distribution.

- $b_j = 0$: centre of the random-effect distribution;
- $b_j > 0$: higher conditional log-odds;
- $b_j < 0$: lower conditional log-odds;
- τ^2 : residual between-LHU heterogeneity.



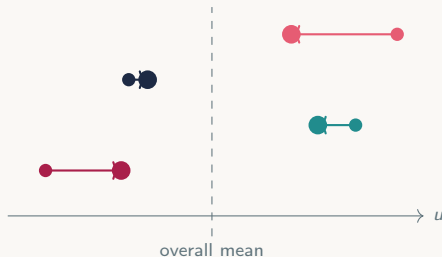
Random effects describe a distribution of clusters; individual $\hat{b}_j$ values are model-based predictions, not ordinary fixed parameters.

Shrinkage: why small LHUs move more

The empirical Bayes/BLUP prediction $\hat{b}_j$ balances:

- the LHU-specific information;
- the estimated between-LHU variance;
- the overall mean.

Sparse or imprecise LHUs are pulled more strongly toward zero. This is a feature, not a defect.



Take-home: Ranking small-area predictions without uncertainty can be misleading.

An approximate ICC for a logistic random-intercept model

Using the latent-variable representation, the level-1 residual variance is $\pi^2/3$:

$$\text{ICC} = \frac{\tau^2}{\tau^2 + \pi^2/3}.$$

- τ standard deviation of random intercept between LHUs.
- It summarises residual similarity of two people in the same LHU on a latent logistic scale.
- It is not a variance decomposition on the observed binary scale.
- A small ICC can still matter when clusters are large.

Example

If $\hat{\tau} = 0.35$, then $\widehat{\text{ICC}} = 0.35^2 / (0.35^2 + \pi^2/3) \approx 0.036$.

Conditional, typical-cluster and marginal predictions

Target	Computation	Question answered
Conditional for LHU j	include $\hat{b}_j$	What is predicted for this observed LHU?
Typical latent LHU	set $b = 0$	What is predicted at the median random-effect value?
Population marginal	average over $b \sim N(0, \hat{\tau}^2)$	What is the average prediction across the LHU distribution?

$$P(Y = 1 \mid \mathbf{x}) = \int \text{logit}^{-1}(\alpha + \mathbf{x}^\top \beta + b) \phi(b; 0, \tau^2) db.$$

Take-home: With a nonlinear link, setting $b = 0$ is not the same as averaging over the random-effect distribution.

GLMM or GEE? Choose by estimand

	GLMM	GEE
Primary target	Cluster-specific association and heterogeneity	Population-averaged association
Dependence	Random-effects distribution	Working correlation
Cluster predictions	Yes, with shrinkage	Not the main goal
Small number of clusters	Model assumptions matter strongly	Sandwich SE may require corrections
Interpretation	Conditional OR	Marginal OR

Neither is universally “better”: the scientific question should decide.

4. Synthetic PASSI example

A cross-sectional analysis of current smoking, clustered by LHU

A synthetic PASSI-like multilevel dataset

The teaching dataset reproduces the main structure of an annual behavioural surveillance sample:

- one synthetic cross-section for 2023;
- define current smoking as a binary outcome;
- treat respondents as nested in LHUs;
- adjust for sex, age, education and economic difficulties;
- stratify individual sampling and attach a final respondent weight.

Model hierarchy

Level 1: respondent

Level 2: LHU

Context: region / macro-area

Take-home: The data-generating process is known, so every step of the workflow can be reproduced.

Research question and model

Question

How do individual characteristics relate to current smoking, and how much residual heterogeneity remains across LHUs?

$$\text{logit}(p_{ij}) = \alpha + \beta_1 \text{Male}_{ij} + \beta_2 \text{Age10}_{ij} + \beta_3^\top \text{Education}_{ij} + \beta_4^\top \text{EconDiff}_{ij} + b_j.$$

This is associational, not causal: exposure and outcome are measured in the same cross-section and confounding may remain.

Analysis variables and coding

Variable	Role	Coding	Role in the synthetic design
smoker	outcome	1 current, 0 not current	generated from a logistic mixed model
lhu	cluster	60 LHU identifiers	level-2 grouping variable
sex	predictor	Female / Male	Female is the reference
age / age10	predictor	18–69; centred per 10 years	continuous individual predictor
education	predictor	three categories	University is the reference
econ_diff	predictor	three categories	None is the reference
weight	design	positive final weight	inverse individual inclusion probability
sampling_stratum	design	four strata	one-stage stratified sampling

Take-home: Even with synthetic data, explicit coding keeps the model and its interpretation aligned.

R setup

```
library(tidyverse)
library(lme4)
library(broom.mixed)
library(performance)
library(DHARMa)
library(survey)
library(ggeffects)
library(sjPlot)

options(contrasts = c("contr.treatment", "contr.poly"))
set.seed(2026)
```

The companion file `GLMM_PASSI_2023_workflow.R` generates the data and runs the complete analysis.

Generate the LHU structure

```
lhu_frame <- tibble(  
  lhu = factor(sprintf("LHU_%02d", 1:60)),  
  region = factor(rep(1:12, each = 5)),  
  n = sample(160:320, 60, replace = TRUE),  
  b_lhu = rnorm(60, mean = 0, sd = 0.35)  
)  
  
dat <- lhu_frame |>  
  tidyr::uncount(n, .id = "respondent")
```

The cluster effect b_j is generated once per LHU and inherited by its respondents.

Generate respondents and the binary outcome

```
dat <- dat |>
  mutate(
    sex = factor(sample(c("Female", "Male"), n(), TRUE)),
    age = sample(18:69, n(), TRUE),
    education = factor(
      sample(c("University", "Secondary", "Lower"), n(),
        TRUE, prob = c(.25, .50, .25)),
      levels = c("University", "Secondary", "Lower")
    ),
    econ_diff = factor(
      sample(c("None", "Some", "Many"), n(),
        TRUE, prob = c(.45, .40, .15)),
      levels = c("None", "Some", "Many")
    ),
    age10 = (age - mean(age)) / 10,
    eta = -1.45 + .45*(sex == "Male") - .10*age10 +
      .25*(education == "Lower") + .30*(econ_diff == "Some") +
      .65*(econ_diff == "Many") + b_lhu,
    smoker = rbinom(n(), 1, plogis(eta))
  )
```

Generate one-stage stratification weights

```
dat <- dat |>
  mutate(
    sampling_stratum = factor(paste(
      sex, if_else(age < 50, "18-49", "50-69"), sep = "_"
    )),
    weight = unname(c(
      "Female_18-49" = 1.35, "Male_18-49" = 1.10,
      "Female_50-69" = 0.90, "Male_50-69" = 1.20
    )[as.character(sampling_stratum)])
  )
```

- Strata are respondent-level sex–age groups, not LHUs.
- The final weight represents one stage of individual selection within strata.
- The LHU is an outcome cluster, but it is not a sampled first-stage PSU in this synthetic design.

Validate the analytic structure

```
dat |> count(smoker)
dat |> count(sex)
summary(dat$age)
dat |> count(education)
dat |> count(econ_diff)

cluster_size <- dat |>
  count(lhu, name = "n")

summary(cluster_size$n)
n_distinct(dat$lhu)
```

Check:

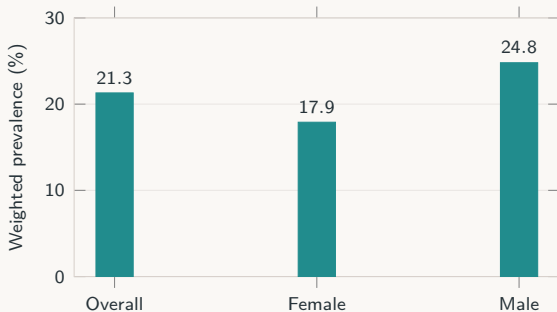
- both outcome levels occur;
- factor references are meaningful;
- enough observations and events exist per LHU;
- every cluster has both covariate variation and outcome information;
- the generated cluster sizes match the intended range.

Design-based descriptive prevalence

```
des <- svydesign(  
  ids = ~1, # one-stage stratified design  
  strata = ~sampling_stratum,  
  weights = ~weight,  
  nest = TRUE,  
  data = dat  
)  
  
svymean(~smoker, des, na.rm = TRUE)  
svyby(~smoker, ~sex, des, svymean, na.rm = TRUE,  
      vartype = c("se", "ci"))
```

- Weighted estimates describe the target population under the declared design.
- Show denominators and uncertainty, not percentages alone.
- The synthetic sample has one stage of individual selection within strata, so no PSU term is required.

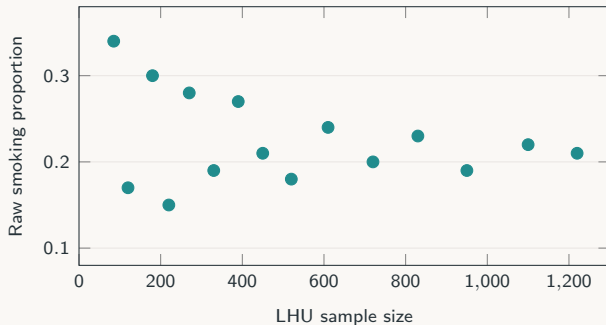
Descriptive smoking prevalence



Synthetic data

The values summarise the generated 2023 teaching sample. The descriptive chart does not adjust for covariates and should not be read as a model result.

Explore LHU heterogeneity before modelling



Synthetic data

Extreme raw proportions often occur in small clusters. A funnel plot or binomial interval helps distinguish sampling noise from plausible heterogeneity.

Start with a single-level logistic model

```
m_glm <- glm(  
  smoker ~ sex + age10 + education + econ_diff,  
  family = binomial(link = "logit"),  
  data = dat  
)  
  
broom::tidy(m_glm, exponentiate = TRUE, conf.int = TRUE)
```

- Useful as a substantive baseline.
- Its default standard errors assume independent observations.
- Comparing coefficients with the GLMM requires care: logistic coefficients change scale when latent heterogeneity is added.

Fit the random-intercept logistic model

```
m_ri <- glmer(  
  smoker ~ sex + age10 + education + econ_diff +  
    (1 | lhu),  
  family = binomial(link = "logit"),  
  data = dat,  
  control = glmerControl(optimizer = "bobyqa")  
)  
  
summary(m_ri)  
performance::check_singularity(m_ri)  
performance::icc(m_ri)
```

Take-home: The model separates the adjusted individual-level associations from residual between-LHU heterogeneity.

Report fixed effects as odds ratios

```
or_tab <- broom.mixed::tidy(  
  m_ri,  
  effects = "fixed",  
  conf.int = TRUE,  
  exponentiate = TRUE  
) |>  
  select(term, estimate, conf.low, conf.high)  
  
or_tab
```

Example sentence template

“Within the same latent LHU context and conditional on the other covariates, the odds of current smoking were [OR] times those in the reference group (95% CI [L, U]).”

Caution: Odds ratios are non-collapsible; a change from the GLM estimate is not, by itself, evidence of confounding.

Packages for model summaries and predictions

Package	Main role	Useful functions
<code>sjPlot</code>	publication-ready tables and model plots	<code>tab_model()</code> , <code>plot_model()</code>
<code>ggeffects</code>	adjusted predictions on the response scale	<code>predict_response()</code> , <code>plot()</code>
<code>performance</code>	model checks and variance summaries	<code>check_model()</code> , <code>check_singularity()</code> , <code>icc()</code>
<code>DHARMa</code>	simulation-based residual diagnostics	<code>simulateResiduals()</code> and formal residual tests

Take-home: Convenience functions do not choose the estimand. Specify whether each prediction is conditional, typical-cluster or population averaged.

A complete model table with sjPlot

```
sjPlot::tab_model(  
  m_ri,  
  transform = "exp",  
  show.ci = 0.95,  
  show.icc = TRUE,  
  show.re.var = TRUE,  
  show.ngroups = TRUE,  
  show.aic = TRUE  
)
```

- Fixed effects appear as odds ratios with confidence intervals.
- The random-effects section reports variance components, ICC and the number of LHUs.
- Add readable coefficient labels before exporting the table for a report.

Fixed and random effects with sjPlot

```
# Fixed effects as conditional odds ratios
sjPlot::plot_model(
  m_ri, type = "est", transform = "exp",
  show.values = TRUE, show.p = FALSE
)

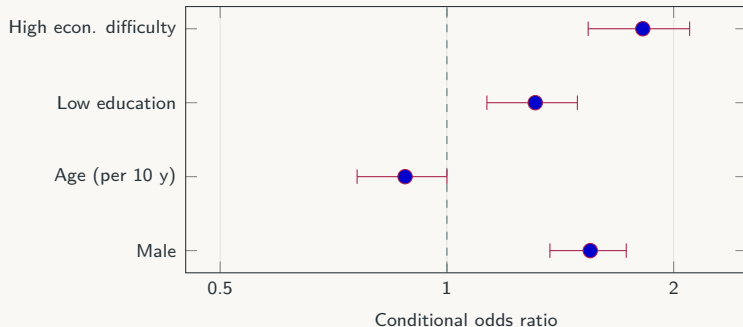
# Predicted LHU deviations with conditional intervals
sjPlot::plot_model(
  m_ri, type = "re", sort.est = TRUE
)
```

The fixed-effect plot uses one as the reference value after exponentiation.

The random-effect plot uses zero as the reference on the log-odds scale.

Caution: Predicted LHU effects include shrinkage and uncertainty; they do not form a league table.

Fixed-effect display



Synthetic model estimates

Plot estimates and 95% CIs on a log scale.
Order terms substantively, not alphabetically.

Quantify LHU heterogeneity

```
vc <- as.data.frame(VarCorr(m_ri))
tau2 <- vc$vcov[vc$grp == "lhu"]
tau  <- sqrt(tau2)

icc_latent <- tau2 / (tau2 + pi^2 / 3)
mor <- exp(0.6744898 * sqrt(2 * tau2))

c(tau2 = tau2, tau = tau,
  ICC = icc_latent, MOR = mor)
```

- τ : SD of LHU intercepts on the log-odds scale.
- ICC: latent-scale clustering.
- Median odds ratio (MOR): median contrast between two randomly chosen LHUs, expressed on the OR scale.

Is the random intercept supported?

```
m_fixed_lhu <- glm(
  smoker ~ sex + age10 + education + econ_diff + lhu,
  family = binomial,
  data = dat
)

# Descriptive model comparison; boundary issues affect naive LRTs.
AIC(m_glm, m_ri)
performance::r2_nakagawa(m_ri)
```

- Do not decide from a single p -value.
- Consider design knowledge, $\hat{\tau}$, its uncertainty, convergence and the number/size of clusters.
- Testing $\tau^2 = 0$ is a boundary problem; a parametric bootstrap is preferable when a formal test is essential.

Extract and display LHU effects

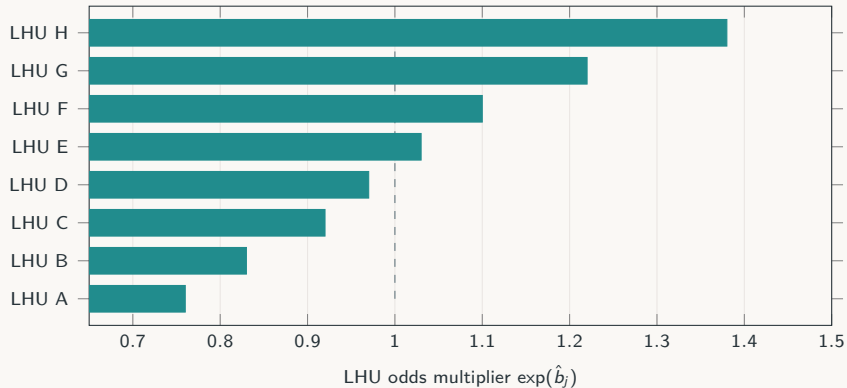
```
re <- ranef(m_ri, condVar = TRUE)$lhu |>
  tibble::rownames_to_column("lhu") |>
  rename(b_hat = '(Intercept)')

re |>
  mutate(odds_multiplier = exp(b_hat)) |>
  arrange(odds_multiplier)
```

Caterpillar plots should include conditional uncertainty intervals and a clear reference at zero (or one after exponentiation).

Caution: They are adjusted model-based predictions, not a definitive league table of LHU quality.

Adjusted LHU heterogeneity in the synthetic data



Synthetic model estimates

Three prediction calls, three meanings

```
newdat <- tidyr::expand_grid(  
  sex = levels(dat$sex),  
  age10 = c(-1.5, 1.5),  
  education = "University",  
  econ_diff = "None",  
  lhu = levels(dat$lhu)[1]  
)  
  
# Conditional for the named observed LHU  
newdat$p_cond <- predict(m_ri, newdat, type = "response")  
  
# b = 0: a typical latent LHU, not a population average  
newdat$p_b0 <- predict(m_ri, newdat, type = "response",  
  re.form = NA)
```

Population-marginal predictions by simulation

```
beta <- fixef(m_ri)
tau  <- as.numeric(attr(VarCorr(m_ri)$lhu, "stddev"))

X <- model.matrix(delete.response(terms(m_ri)), newdat)
b_draw <- rnorm(50000, mean = 0, sd = tau)

newdat$p_marginal <- apply(X, 1, function(xrow) {
  mean(plogis(drop(xrow %*% beta) + b_draw))
})
```

For publication, propagate uncertainty in both β and τ using parametric bootstrap or posterior simulation, rather than averaging only over b .

Adjusted predictions withggeffects

```
# Typical LHU: population-level prediction with  $b = 0$ 
p_typical <- ggeffects::predict_response(
  m_ri,
  terms = c("age10 [-2:2 by=.5]", "sex")
)

# Average predictions over the observed covariate distribution
p_average <- ggeffects::predict_response(
  m_ri,
  terms = c("age10 [-2:2 by=.5]", "sex"),
  margin = "empirical"
)

plot(p_typical)
plot(p_average)
```

The default population-level prediction describes a typical LHU. With `margin = "empirical"`, the package averages counterfactual predictions over the observed sample.

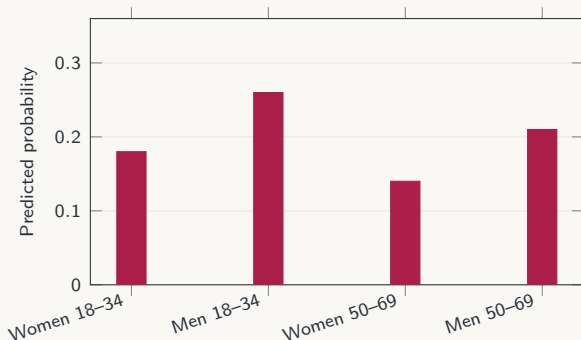
Unit-level predictions and survey-weighted averaging

```
# Conditional predictions for selected observed LHUs
p_lhu <- ggeffects::predict_response(
  m_ri,
  terms = c("age10 [-1.5,1.5]",
            "lhu [LHU_01,LHU_02]"),
  type = "random"
)

# Weight the empirical averaging step
p_average_w <- ggeffects::predict_response(
  m_ri,
  terms = "sex",
  margin = "empirical",
  weights = "weight"
)
```

- Unit-level predictions include the estimated random effects of the named LHUs.
- The `weights` argument changes how predictions are averaged over rows.
- It does not convert an unweighted GLMM into a design-based multilevel estimator.

Predicted probabilities communicate absolute differences



Synthetic marginal predictions

State what is held fixed or averaged over.

Add uncertainty bands when presenting the final model results.

One-stage stratified sampling

Synthetic design

Individuals are sampled within strata. There is no additional sampling stage.

Final weight

$$w_{ij} = \frac{1}{\Pr(i \text{ selected} \mid h)}$$

is a respondent-level weight for stratum h .

- For design-based descriptives, declare `ids = ~1`, strata and final weights.
- The LHU random effect describes outcome dependence; the stratum describes sampling.
- Include stratification variables in the outcome model when they are substantively relevant.
- A weighted GLMM still requires a method with a sampling-weight interpretation; `glmer(weights=)` uses prior weights.

One-stage weights: a transparent sensitivity analysis

```
dat <- dat |>
  group_by(lhu) |>
  mutate(weight_l1_scaled = weight / mean(weight)) |>
  ungroup()

m_ri_weighted <- update(
  m_ri,
  data = dat,
  weights = weight_l1_scaled
)
```

- Scaling within LHU preserves an average level-1 weight of one.
- Compare fixed effects and $\hat{\tau}^2$ with the unweighted GLMM.
- In lme4, these are prior weights: treat this fit as a sensitivity analysis, not as a complete design-based multilevel estimator.

A multistage design requires level-specific weights

Suppose LHUs are sampled first and individuals are sampled within selected LHUs:

$$\pi_{ij} = \pi_j \pi_{i|j}, \quad w_{ij}^{\text{final}} = \frac{1}{\pi_{ij}} = \underbrace{\frac{1}{\pi_j}}_{w_j} \times \underbrace{\frac{1}{\pi_{i|j}}}_{w_{i|j}}.$$

- The final weight combines the two sampling stages.
- A weighted multilevel analysis may need the cluster weight w_j and the conditional individual weight $w_{i|j}$ separately.
- Do not reuse the same combined weight as if it represented both levels.
- Scaling must be defined within levels and documented.

Caution: For multistage samples, use a multilevel pseudo-likelihood or Bayesian method that supports level-specific weights, and run sensitivity analyses.

5. Diagnostics and extensions

Make the model earn its interpretation

A diagnostic workflow

```
# Optimisation messages
m_ri@optinfo$conv$lme4$messages

# Random-effects structure
lme4::isSingular(m_ri)
performance::check_singularity(m_ri)

# Fixed part
performance::check_collinearity(m_ri)
```

```
# Combined visual checks
performance::check_model(m_ri)

# Binomial dispersion
performance::check_overdispersion(m_ri)

# Compare optimizers if needed
af <- lme4::allFit(m_ri)
summary(af)
```

Begin with convergence and identifiability, then examine residual behaviour and sensitivity to modelling choices.

Simulation-based residual checks

```
set.seed(2026)
res <- DHARMA::simulateResiduals(
  fittedModel = m_ri,
  n = 1000
)

plot(res)
DHARMA::testUniformity(res)
DHARMA::testDispersion(res)
DHARMA::testOutliers(res)
```

- Inspect residual patterns against fitted values and key predictors.
- A binary model cannot be diagnosed by a normal Q–Q plot of raw residuals.
- Domain knowledge remains essential: diagnostics do not detect every form of misspecification.

A convergence warning is evidence, not decoration

1. Re-check coding, separation, sparse cells and cluster sizes.
2. Centre/scale continuous predictors and simplify redundant terms.
3. Fit with a well-supported optimizer and adequate iterations.
4. Compare optimizers with `allFit()` where useful.
5. Inspect gradient, Hessian and singularity; do not merely suppress warnings.
6. If the model is not supported by the data, simplify it and report the limitation.

Caution: Changing the optimizer cannot rescue an unidentified or substantively incoherent model.

Common failure modes in this example

Symptom	Likely cause	Response
Singular fit	random-effect variance near zero or over-complex structure	simplify; report estimate and uncertainty
Huge coefficients/SEs	complete or quasi-separation	inspect tables; collapse only if substantive; consider penalisation
Unstable LHU effects	few events or respondents in clusters	show uncertainty; avoid rankings
Residual pattern	omitted nonlinearity or dependence	revise fixed/random structure
Weight-sensitive results	informative sampling or scaling choices	document design; run sensitivity analyses

Numerical example: the same fixed effects, different LHUs

Assume $\alpha = -1.20$ and $\beta = -0.15$ for age measured in ten-year units.

LHU	b_{0j}	b_{1j}	intercept	age slope	age OR
Typical	0	0	-1.20	-0.15	$e^{-0.15} = 0.86$
A	0.40	0.10	-0.80	-0.05	$e^{-0.05} = 0.95$
B	-0.30	-0.20	-1.50	-0.35	$e^{-0.35} = 0.70$

At the centred age, the smoking probabilities are

$$\text{logit}^{-1}(-1.20) = 0.23, \quad \text{logit}^{-1}(-0.80) = 0.31, \quad \text{logit}^{-1}(-1.50) = 0.18.$$

- LHU A has a higher baseline and a nearly flat age association.
- LHU B has a lower baseline and a stronger negative age association.

The mapping to lme4 formulas

Random structure	Formula component	Meaning
Random intercept	$(1 \mid \text{1hu})$	b_{0j} varies
Random slope only	$(0 + \text{age10} \mid \text{1hu})$	b_{1j} varies with a common intercept at $\text{age10} = 0$
Correlated intercept and slope	$(1 + \text{age10} \mid \text{1hu})$	b_{0j} and b_{1j} vary with an estimated covariance
Uncorrelated intercept and slope	$(1 \mid \text{1hu}) + (0 + \text{age10} \mid \text{1hu})$	both vary with covariance fixed at zero

```
# Keep the population-average age slope in the fixed part  
smoker ~ age10 + (1 + age10 | 1hu)
```

The fixed term age10 estimates β . The random term estimates the distribution of deviations b_{1j} around that fixed slope.

What fixef, ranef and coef return

```
fixef(m_slope)
# alpha and beta: centre of the LHU distribution

head(ranef(m_slope)$lhu)
# b_0j and b_1j: predicted LHU deviations

head(coef(m_slope)$lhu)
# alpha + b_0j and beta + b_1j: LHU coefficients

VarCorr(m_slope)
# tau_0, tau_1 and their correlation
```

Reading one row of `coef(m_slope)$lhu`

The intercept column gives the fitted baseline log-odds for that LHU. The age10 column gives its fitted ten-year age slope.

The values from `ranef()` are shrinkage predictions of random-effect realizations, rather than a new set of unrelated fixed parameters.

Centering determines the meaning of the random intercept

With $\text{age10} = (\text{age} - 45)/10$, $x = 0$ corresponds to age 45.

Intercept

$\alpha + b_{0j}$ describes the baseline log-odds at age 45 for the reference categories.

Intercept-slope correlation

The correlation asks whether LHUs with higher smoking odds at age 45 tend to have stronger or weaker age gradients.

Changing the centering point changes the numerical intercept and often changes the estimated intercept-slope correlation. The fitted probabilities remain the same after a correct reparameterization.

Take-home: Center continuous predictors at a value that makes the cluster-specific intercept scientifically interpretable.

Random slopes: when an association varies by LHU

Suppose the age association may differ across LHUs:

```
m_slope <- glmer(  
  smoker ~ sex + age10 + education + econ_diff +  
    (1 + age10 | lhu),  
  family = binomial,  
  data = dat,  
  control = glmerControl(optimizer = "bobyqa")  
)  
  
summary(m_slope)  
performance::check_singularity(m_slope)
```

- Requires enough age variation and events within many LHUs.
- Estimate and interpret intercept variance, slope variance and their correlation.
- Complexity should follow the design and scientific question, not habit.

Evidence for a random slope

Scientific requirement

The question must allow the age association to differ across LHUs. A random slope estimates that heterogeneity.

Information in the data

Age must vary within LHUs, with enough respondents and smoking events across the age range.

Examine $\hat{\tau}_1$, its uncertainty, the intercept-slope correlation, singularity and stability across optimizers. The likelihood-ratio reference distribution requires care because variance components lie on the boundary under the simpler model.

```
VarCorr(m_slope)
performance::check_singularity(m_slope)
```

```
AIC(m_ri, m_slope)
anova(m_ri, m_slope,
      test = "Chisq")
```

An additional random intercept for region

Respondents are nested in LHUs, and LHUs are nested in regions:

```
m_threelevel <- glmer(
  smoker ~ sex + age10 + education + econ_diff +
    (1 | region) + (1 | lhu),
  family = binomial,
  data = dat,
  control = glmerControl(optimizer = "bobyqa")
)

sjPlot::plot_model(m_threelevel, type = "re")
performance::icc(m_threelevel, by_group = TRUE)
```

- The model allocates residual heterogeneity to the region and LHU levels.
- Twelve regions provide limited information for estimating a region variance, so uncertainty and singularity require close attention.
- Use this structure only when the hierarchy matches the sampling and substantive context.

Contextual predictors separate composition from context

Add an LHU-level covariate, for example an area deprivation score z_j :

```
m_context <- glmer(  
  smoker ~ sex + age10 + education + econ_diff +  
    lhu_deprivation + (1 | lhu),  
  family = binomial,  
  data = dat  
)
```

- Individual economic difficulty and LHU deprivation answer different questions.
- Centre contextual predictors to make the intercept meaningful.
- If an individual exposure has both within- and between-LHU variation, consider group-mean decomposition to avoid conflating levels.

What a cross-sectional PASSI GLMM can and cannot say

It can estimate

- adjusted associations;
- residual LHU heterogeneity;
- conditional and marginal predictions;
- shrinkage-adjusted LHU contrasts.

It cannot establish

- temporal order;
- causal effects without stronger assumptions;
- change over time from one annual cross-section;
- performance rankings free of measurement error.

6. Guided exercise

Thirty minutes: fit, interpret, diagnose, communicate (not mandatory)

Exercise brief

1. Inspect outcome and cluster sizes.
2. Fit the ordinary logistic model and the random-intercept GLMM.
3. Report one adjusted OR with a conditional interpretation.
4. Compute $\hat{\tau}$, latent ICC and MOR.
5. Fit the age random-slope model and interpret one LHU-specific slope.
6. Produce one predicted probability contrast and label its target.
7. Run one simulation-based diagnostic and state what it checks.

Debrief prompts

1. Did the GLM and GLMM coefficients differ? Why might that happen?
2. Was the LHU variance substantively meaningful even if the ICC was small?
3. Which prediction target did you choose, and why?
4. Which clusters were most uncertain? Was shrinkage visible?
5. What would you change before using the analysis in a report?

Take-home: A defensible answer states the estimand, acknowledges the design and makes uncertainty visible.

A compact reporting checklist

- annual sample and exclusions;
- outcome definition and references;
- cluster unit and number of clusters;
- fixed and random structure;
- estimation software and version;
- convergence/singularity checks;
- ORs with 95% CIs;
- τ^2 plus ICC and/or MOR;
- prediction target and averaging;
- treatment of survey design/weights;
- diagnostics and sensitivity analyses;
- cross-sectional and measurement limits.

Seven messages to take away

1. Clustering changes uncertainty and can be scientifically meaningful.
2. Fixed effects describe the centre of the cluster distribution.
3. A random intercept changes the cluster baseline, whereas a random slope changes a cluster association.
4. Cluster coefficients equal fixed effects plus their random deviations.
5. Partial pooling stabilises cluster predictions.
6. Conditional, $b = 0$ and marginal probabilities are different targets.
7. Diagnostics must support the chosen random-effects structure.

Questions?

Appendix

Theory retained from the original slide pack and reusable R patterns

A.1 The marginal likelihood requires integration

Conditionally on u_j , responses are independent. The cluster contribution is

$$L_j(\beta, \tau) = \int \left\{ \prod_{i=1}^{n_j} p_{ij}(u)^{y_{ij}} [1 - p_{ij}(u)]^{1-y_{ij}} \right\} \phi(u; 0, \tau^2) du.$$

The full likelihood is $L = \prod_j L_j$.

- For non-Gaussian outcomes, the integral usually has no closed form.
- Software uses approximations such as Laplace or adaptive Gauss–Hermite quadrature.
- Accuracy depends on cluster information and the complexity of the random-effects structure.

A.2 Matrix notation and induced covariance

For a Gaussian mixed model,

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \varepsilon, \quad \mathbf{u} \sim N(\mathbf{0}, \mathbf{G}), \quad \varepsilon \sim N(\mathbf{0}, \mathbf{R}).$$

Therefore,

$$E(\mathbf{Y}) = \mathbf{X}\beta, \quad \text{Var}(\mathbf{Y}) = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R}.$$

For a random intercept, rows belonging to the same cluster share the same column of $\mathbf{Z}$.

The shared random component creates non-zero within-cluster covariance after $\mathbf{u}$ is integrated out.

The GLMM keeps this structural idea, but models the conditional mean through a link function.

A.3 Random intercept–slope covariance

With a random intercept and a random slope u_{1j} ,

$$\text{logit}(p_{ij}) = \beta_0 + \beta_1 x_{ij} + u_{0j} + u_{1j} x_{ij}, \quad \begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \rho \tau_0 \tau_1 \\ \rho \tau_0 \tau_1 & \tau_1^2 \end{pmatrix} \right].$$

- τ_0^2 : baseline heterogeneity;
- τ_1^2 : heterogeneity in the x association;
- ρ : association between cluster baseline and cluster-specific slope.

A boundary estimate or correlation near ± 1 often signals that the data do not support the full structure.

A.4 Common lme4 formula patterns

```
# Random intercept
y ~ x1 + x2 + (1 | cluster)

# Correlated random intercept and slope
y ~ x1 + x2 + (1 + x1 | cluster)

# Uncorrelated random intercept and slope
y ~ x1 + x2 + (1 | cluster) + (0 + x1 | cluster)

# Respondents nested in LHU nested in region
y ~ x1 + x2 + (1 | region/lhu)

# Crossed grouping factors
y ~ x1 + x2 + (1 | interviewer) + (1 | lhu)
```

Parentheses describe the random-effects structure; nesting in the data must match the formula, not merely its label.

A.5 Other outcome families

Outcome	Family	Common link	Example
Continuous	Gaussian	identity	health score
Binary	Binomial	logit	current smoker
Count	Poisson	log	number of visits
Overdispersed count	Negative binomial	log	recurrent events
Proportion with denominator	Binomial	logit	successes / trials
Ordinal	Cumulative mixed model	link logit/probit	ordered health status

The outcome distribution and mean–variance relation must be justified; a random effect does not repair the wrong family.

A.6 About the subject-specific to marginal approximation

The original material included an approximate attenuation relationship of the form

$$\beta^M \approx \frac{\beta^{SS}}{\sqrt{1 + c\tau^2}},$$

where the constant c depends on the approximation used.

- It is useful for intuition: marginal log-odds coefficients are typically closer to zero.
- It is not a universal identity and does not replace explicit marginalisation.
- For teaching and reporting, compute marginal predictions or contrasts directly from the fitted model.

A.7 References and further reading

- Agresti, A. (2013). *Categorical Data Analysis*, 3rd ed. Wiley.
- Bates, D., Maechler, M., Bolker, B., Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1).
- Bolker, B. et al. (2009). Generalized linear mixed models: a practical guide for ecology and evolution. *Trends in Ecology & Evolution*, 24(3).
- McCulloch, C., Searle, S., Neuhaus, J. (2008). *Generalized, Linear, and Mixed Models*, 2nd ed. Wiley.
- Merlo, J. et al. (2006). A brief conceptual tutorial of multilevel analysis in social epidemiology. *Journal of Epidemiology & Community Health*.
- PASSI documentation and annual metadata: <https://www.epicentro.iss.it/passi/>

A.8 Software documentation

- lme4: <https://cran.r-project.org/package=lme4>
- sjPlot: <https://strengejacke.github.io/sjPlot/>
- ggeffects: <https://strengejacke.github.io/ggeffects/>
- performance: <https://easystats.github.io/performance/>
- DHARMA: <https://cran.r-project.org/package=DHARMA>

The examples in the main deck use the current `predict_response()` interface. For mixed models, the default gives a typical-cluster prediction, `margin = "empirical"` averages over the observed sample, and `type = "random"` requests predictions for named cluster levels.

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